

The Outer-Phase Identity for the Warped Pullback Spectrum

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Abstract

Let $H(x) := Z(\theta^{-1}(x))/\sqrt{\theta'(\theta^{-1}(x))}$ be the warped pullback of the Hardy Z -function, with θ the Riemann–Siegel phase. By Theorem B of the companion document (“Three Missing Theorems”), the windowed modulus $|\hat{H}_T(\xi)|^2/(\theta(T) - X_0)$ converges to a nonnegative, compactly-supported spectral density $S(\xi)$ with explicit mode-sum form on $(-1, 1)$, and the whitened differential factorizes as $d\hat{H} = \sqrt{S(\xi)} e^{i\Phi(\xi)} d\xi$ in $\mathcal{S}'([-1 + \delta, 1 - \delta])$ with $\Phi(\xi) = \arg \sum_{n,\sigma} |\mathcal{A}_{n,\sigma}(\xi)| e^{i\phi_{n,\sigma}(\xi)}$. Define the analytic signal $H_+(z) := \int_0^1 e^{i\xi z} d\hat{H}(\xi)$. We reduce the question “ H_+ is outer in $\mathbb{C}^+$ ” to a single explicit identity on $(-1 + \delta, 1 - \delta)$:

$$\Phi(\xi) \equiv \mathcal{H}[\tfrac{1}{2} \log S](\xi) \pmod{\pi},$$

where $\mathcal{H}$ is the Hilbert transform on $(-1, 1)$. This is the Krein–Szegő outer-phase condition specialized to the explicit phase produced by Theorem B.

1 Inputs

Definition 1 (Warped pullback). $H(x) := Z(\theta^{-1}(x))/\sqrt{\theta'(\theta^{-1}(x))}$, $x \geq X_0 := \theta(T_0)$, $T_0 \geq 200$.

Definition 2 (Windowed Fourier transform). $\hat{H}_T(\xi) := \int_{X_0}^{\theta(T)} H(x) e^{-i\xi x} dx$.

Inputs from prior results (cited from companion documents). (**Band-limit**, `BandLimitedness.tex`, **Theorem 9**). $|\hat{H}_T(\xi)|^2/(\theta(T) - X_0) \rightarrow 0$ for every fixed $|\xi| > 1$, locally uniformly.

(**Theorem B of three_missing_theorems**). For almost every $\xi \in (-1, 1)$,

$$\lim_{T \rightarrow \infty} \frac{|\hat{H}_T(\xi)|^2}{\theta(T) - X_0} = S(\xi) := \frac{1}{(2\pi)^2} \sum_{n \geq 1, \sigma \in \{\pm\}} |\mathcal{A}_{n,\sigma}(\xi)|^2, \quad (1)$$

with $\mathcal{A}_{n,\sigma}(\xi) = n^{-1/2} \sqrt{2\pi/|\theta''(t_{n,\sigma,\xi}^*)|} e^{i\pi/4}$ the stationary-phase amplitude and $t_{n,\sigma,\xi}^* := (\theta')^{-1}(\sigma \log n/(\sigma - \xi))$. Moreover, on $\mathcal{S}'([-1 + \delta, 1 - \delta])$,

$$d\hat{H}(\xi) = \sqrt{S(\xi)} e^{i\Phi(\xi)} d\xi, \quad \Phi(\xi) := \arg \sum_{n,\sigma} |\mathcal{A}_{n,\sigma}(\xi)| e^{i\phi_{n,\sigma}(\xi)}, \quad (2)$$

where $\phi_{n,\sigma}(\xi)$ is the stationary-phase phase of mode (n, σ) .

2 The analytic signal and its $H^\infty(\mathbb{C}^+)$ factorization

Definition 3 (Analytic signal).

$$H_+(z) := 2 \int_{[0,1]} e^{i\xi z} d\hat{H}(\xi) = 2 \int_0^1 e^{i\xi z} \sqrt{S(\xi)} e^{i\Phi(\xi)} d\xi, \quad z \in \mathbb{C}.$$

Lemma 4 (Basic properties). H_+ is entire; $|H_+(z)| \leq 2 \int_0^1 \sqrt{S(\xi)} d\xi < \infty$ on $\overline{\mathbb{C}^+}$; $|H_+(iy)| \rightarrow 0$ as $y \rightarrow +\infty$; on $\mathbb{R}$, $H_+(u) = H(u) + i\mathcal{H}[H](u)$.

Proof. Absolute convergence of the integral for all $z \in \mathbb{C}$ from compact support of the integrand. Bound in $\mathbb{C}^+$ by $|e^{i\xi z}| = e^{-\xi \operatorname{Im} z} \leq 1$ for $\xi \geq 0$, $\operatorname{Im} z \geq 0$. Decay on $i\mathbb{R}^+$ by dominated convergence: $e^{-\xi y} \rightarrow 0$ on $(0, 1]$. Reality identity from $\widehat{H}_+ = 2\mathbf{1}_{[0,1]} \widehat{H}$ combined with $\widehat{H}(-\xi) = \overline{\widehat{H}(\xi)}$. $\square$

Lemma 5 ($H^\infty(\mathbb{C}^+)$ canonical factorization). $H_+ \in H^\infty(\mathbb{C}^+)$ admits the canonical factorization $H_+ = B \cdot S_\infty \cdot F$ in $\mathbb{C}^+$, with B a Blaschke product over zeros of H_+ in $\mathbb{C}^+$, S_∞ a singular inner factor supported at infinity, and F outer. Since H_+ is entire with boundary values controlled by Lemma 4, $S_\infty \equiv 1$ (no singular inner at infinity: Lemma 4 gives sub-exponential decay of $\log |H_+(iy)|$ relative to y). Hence

$$H_+(z) = B(z) F(z), \quad z \in \mathbb{C}^+.$$

Proof. Standard canonical factorization for H^∞ of the upper half-plane (Garnett, *Bounded Analytic Functions*, Ch. II, Thm. 4.6). Triviality of S_∞ : by Lemma 4, $|H_+(iy)| \rightarrow 0$ but without exponential rate; indeed $\log |H_+(iy)|/y \rightarrow 0$ as $y \rightarrow \infty$, which is the criterion for trivial singular inner factor at infinity for half-plane H^∞ . $\square$

Corollary 6. $H_+ \in$ Hermite–Biehler class on $\mathbb{C}^+$ (i.e., H_+ has no zeros in $\mathbb{C}^+$) if and only if $B \equiv 1$, i.e., $H_+ = F$ is outer in $\mathbb{C}^+$.

3 Krein–Szegő outer function from the spectral density

Definition 7 (Szegő outer function of S on the band). Let $\delta \in (0, 1)$ small. For $\xi \in (-1 + \delta, 1 - \delta)$, assume $S(\xi) > 0$ (positivity of the spectral density on the interior, from Theorem B mode-sum structure). Define

$$F_S(z) := \exp \left(\frac{1}{2\pi i} \int_{-1+\delta}^{1-\delta} \frac{1+\xi z}{\xi-z} \cdot \frac{\log S(\xi)}{1+\xi^2} d\xi \right), \quad z \in \mathbb{C}^+.$$

This is the Krein–Szegő outer function with $|F_S(\xi)|^2 = S(\xi)$ a.e. on $(-1 + \delta, 1 - \delta)$ and $F_S \in H^2(\mathbb{C}^+)$ outer.

Lemma 8 (Boundary phase of F_S). The boundary values of F_S on $(-1 + \delta, 1 - \delta)$ satisfy

$$F_S(\xi + i0^+) = \sqrt{S(\xi)} \cdot e^{i\Psi(\xi)}, \quad \Psi(\xi) := \mathcal{H}_\delta \left[\frac{1}{2} \log S \right] (\xi),$$

where $\mathcal{H}_\delta$ is the Hilbert transform on $(-1 + \delta, 1 - \delta)$ with kernel $(1 + \xi\eta)/((\xi - \eta)(1 + \eta^2))$ (the finite-interval Plemelj kernel corresponding to the Schwarz integral).

Proof. Apply the Plemelj formula to the Schwarz integral defining $\log F_S$. The Schwarz kernel $(1 + \xi z)/(\xi - z) \cdot 1/(1 + \xi^2)$ decomposes on the real axis boundary as $\pi i \delta(\xi - \eta) + \text{p.v.} (1 + \xi\eta)/((\xi - \eta)(1 + \eta^2))$. Taking Re and Im of $\log F_S(\eta + i0^+)$ gives $\operatorname{Re} \log F_S = \frac{1}{2} \log S$ and $\operatorname{Im} \log F_S = \mathcal{H}_\delta \left[\frac{1}{2} \log S \right]$. $\square$

4 The outer-phase identity

Theorem 9 (Outer-phase identity). H_+ is outer in $\mathbb{C}^+$ if and only if

$$\Phi(\xi) \equiv \mathcal{H}_\delta \left[\frac{1}{2} \log S \right] (\xi) \pmod{\pi}, \quad \text{a.e. } \xi \in (-1 + \delta, 1 - \delta), \quad (3)$$

for every $\delta \in (0, 1)$, where Φ is the whitened phase of (2) and $\mathcal{H}_\delta$ is the Hilbert transform of Lemma 8.

Proof. Two boundary-value functions of $H^2(\mathbb{C}^+)$ functions agree as boundary values (up to a sign in the boundary argument) iff the functions agree up to sign. By Lemma 4 and (2), H_+ has boundary values on $\mathbb{R}$ whose restriction to $(-1 + \delta, 1 - \delta)$ in the $d\hat{H}$ -sense is $\sqrt{S(\xi)} e^{i\Phi(\xi)} d\xi$. The Krein–Szegő outer function F_S has boundary values $\sqrt{S(\xi)} e^{i\Psi(\xi)} d\xi$ (Lemma 8) with $\Psi = \mathcal{H}_\delta \left[\frac{1}{2} \log S \right]$.

Forward. If H_+ is outer, then $B \equiv 1$ in Lemma 5, so $H_+ = F$ coincides with the outer function having $|F(\xi)|^2 = S(\xi)$ a.e., which is F_S up to a unimodular constant. Matching boundary phases gives (3).

Converse. Suppose (3) holds. Then on $(-1 + \delta, 1 - \delta)$, $d\hat{H}(\xi) = \sqrt{S(\xi)} e^{i\mathcal{H}_\delta \left[\frac{1}{2} \log S \right] (\xi)} d\xi$ equals the boundary value of $F_S \cdot d\xi$. Since $\delta > 0$ is arbitrary, the identity extends to $(-1, 1)$ a.e. Hence

$$H_+(z) = 2 \int_0^1 e^{i\xi z} \sqrt{S(\xi)} e^{i\mathcal{H}_\delta \left[\frac{1}{2} \log S \right] (\xi)} d\xi = 2F_S(z) \quad (\text{up to sign / unimodular constant}),$$

which is outer in $\mathbb{C}^+$ by construction. □

Corollary 10. H has only real zeros (Riemann Hypothesis in warped coordinates) if and only if the identity (3) holds.

Proof. H has only real zeros $\iff H_+$ is outer in $\mathbb{C}^+$ (Hermite–Biehler: Corollary 6 combined with the symmetrization $\tilde{H} = \frac{1}{2}(H_+ + H_+^*)$ having zeros exactly at real zeros of H) $\iff$ (3) (Theorem 9). □

5 What (3) says concretely

Both sides of (3) are explicit functions on $(-1 + \delta, 1 - \delta)$:

LHS. $\Phi(\xi) = \arg \sum_{n \geq 1, \sigma \in \{\pm\}} |\mathcal{A}_{n,\sigma}(\xi)| e^{i\phi_{n,\sigma}(\xi)}$. Both $|\mathcal{A}_{n,\sigma}(\xi)|$ and $\phi_{n,\sigma}(\xi)$ are explicit functions of ξ through the stationary-phase locus $t_{n,\sigma,\xi}^*$ and the Riemann–Siegel phase θ . For mode $(n, +)$: $t_{n,+, \xi}^* = (\theta')^{-1}(\log n / (1 - \xi))$, $|\mathcal{A}_{n,+}(\xi)|^2 = 2\pi \log n / (n(1 - \xi)^2 \theta''(t_{n,+, \xi}^*))$, and $\phi_{n,+}(\xi) = (1 - \xi)\theta(t_{n,+, \xi}^*) - t_{n,+, \xi}^* \log n - \pi/4$.

RHS. $\mathcal{H}_\delta \left[\frac{1}{2} \log S \right]$ with S the same mode-sum of squared amplitudes divided by $(2\pi)^2$.

The identity (3) says: *the argument of the mode-sum $\sum |\mathcal{A}_{n,\sigma}| e^{i\phi_{n,\sigma}}$ equals the Hilbert transform of one-half the logarithm of the sum of squared amplitudes.* This is a nontrivial identity between two computable functions.

6 Status

Theorem 9 is a clean reformulation: the “only real zeros” question for H becomes the single-identity question (3). Neither direction is proved in this document. The LHS and RHS are both fully explicit from Theorem B and the Krein–Szegő apparatus. What remains is:

1. Verification of (3) on numerical intervals (computable to any precision from the mode-sum formulas; provides evidence but not proof).
2. Analytic proof of (3) as an identity of Fourier-analytic objects, exploiting the Riemann–Siegel structure.

The identity (3) is the target. It is what must be proven to close the outer-in- $\mathbb{C}^+$ route.

References

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